

Week 6: Propensity Score Weighting

Overview

This practice demonstrates the use of the `PSweight` package in R to estimate causal effects using propensity score weighting with machine learning method, such as Boosted Trees and Super-Learner that integrates multiple ML methods. Augmented IPW is also introduced. We use the NCDS (National Child Development Survey) dataset included in the `PSweight` package. If you are interested in the full use of the package, read the [tutorial](#).

Treatment: `Dany` (indicator of any further education beyond age 16), Outcome: `wage` (log wage) Covariates: parental education, parental employment, ethnicity, etc.

```
library(PSweight)
library(SuperLearner)
library(ggplot2)

## load NCDS dataset
data("NCDS", package="PSweight")
head(NCDS)
```

	white	wage	Dany	Dmult	maemp	scht	qmab	qmab2	qvab	qvab2	paed_u	maed_u
1	1	2.569628	1	>=A/eq	0	2	2	2	1	2	9	9
2	1	2.038615	1	>=A/eq	0	1	5	5	5	5	0	0
3	1	1.722387	0	None	0	1	4	4	4	5	0	0
4	1	2.204874	1	0/eq	0	3	5	4	4	5	10	10
5	1	2.477379	1	0/eq	1	1	3	3	2	3	9	9
6	1	1.502993	0	>=A/eq	1	2	1	1	2	2	10	9

	agepa	agemu	sib_u	wagebin
1	60	59	3	1
2	56	56	0	0
3	57	53	0	0
4	40	41	1	1

5	57	45	1	1
6	43	42	1	0

Setup

Define treatment, outcome, covariates, and propensity score and regression equation. Note that the missing function `ftilt()` in `class` will no longer be missing if you specify the `weight` parameter correctly as shown below.

```
## covariates
covars <- c("white", "maemp", "qmab2", "paed_u", "maed_u", "agepa")

## define propensity score
ps.formula <- reformulate(covars, response = "Dany")
```

Estimate Treatment Effect by Propensity Score Weighting Using Boosted Trees

```
res_gbm <- PSweight(
  ps.formula = ps.formula,
  yname = "wage",
  data = NCDS,
  weight = "IPW",          ## estimand; it can be `IPW` for estimating ATE, `treated` for AT
  ps.method = "gbm",       ## propensity fit by gradient boosting (gbm) or SuperLearner; of
  ps.control = list(n.trees=500, interaction.depth=3)
)

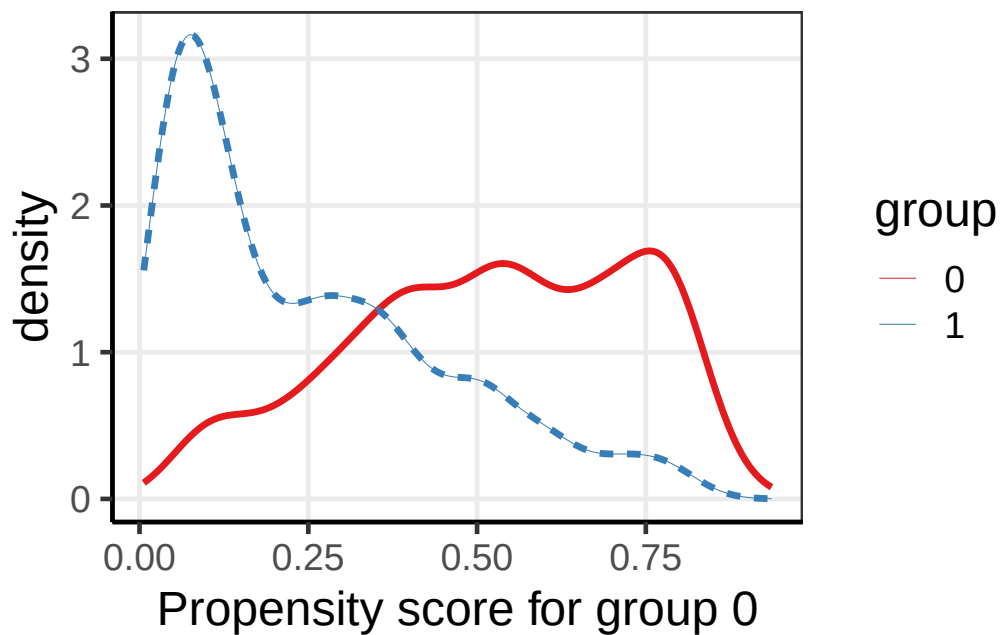
## summarize results
summary(res_gbm)$estimates
```

		Estimate	Std.Error	z value	lwr	upr	Pr(> z)
Contrast	1	0.2448626	0.02177995	11.24257	0.2021747	0.2875505	0

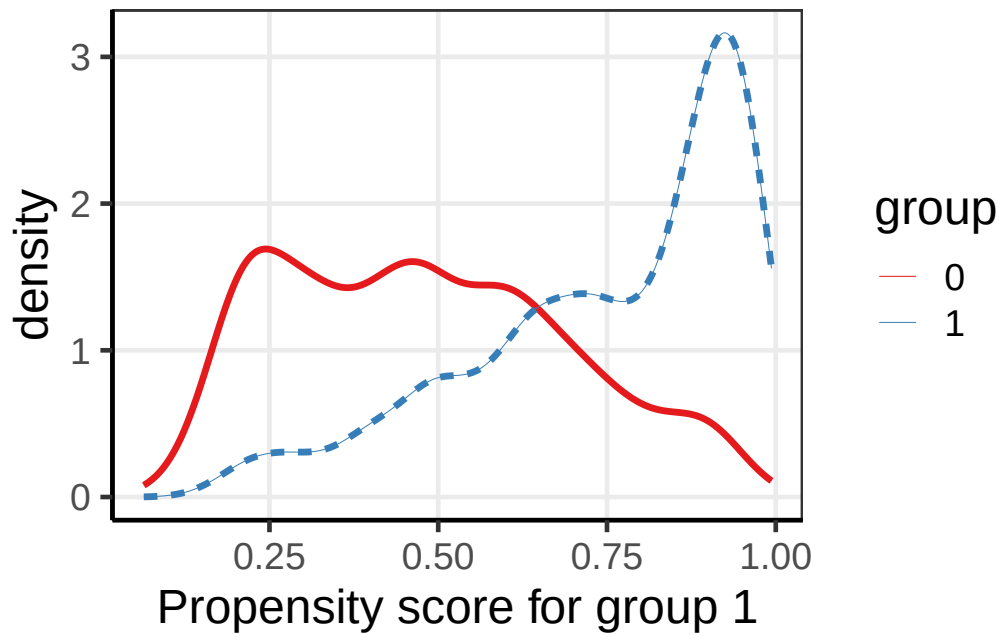
One important step after estimating the propensity score is to evaluate the common support of propensity across treated and control groups, to ensure that there is sufficient overlap in the distribution of estimated scores. This overlap indicates that for each treated unit, there exist comparable control units (and vice versa), allowing for meaningful causal inference without extrapolation beyond observed data.

```
## PSweight has a built in function for checking common support and covariate balance
bal.mult <- SumStat(
  ps.formula = ps.formula,
  data = NCDS,
  weight = c("IPW", "overlap", "entropy", "treated"), ## various approaches to weight individuals
  method = "gbm", ## propensity fit by gradient boosting (gbm) or SuperLearner
  ps.control = list(n.trees=500, interaction.depth=3)
)

## plot the propensity score density
plot(bal.mult, type = "density")
```



Press [enter] to continue

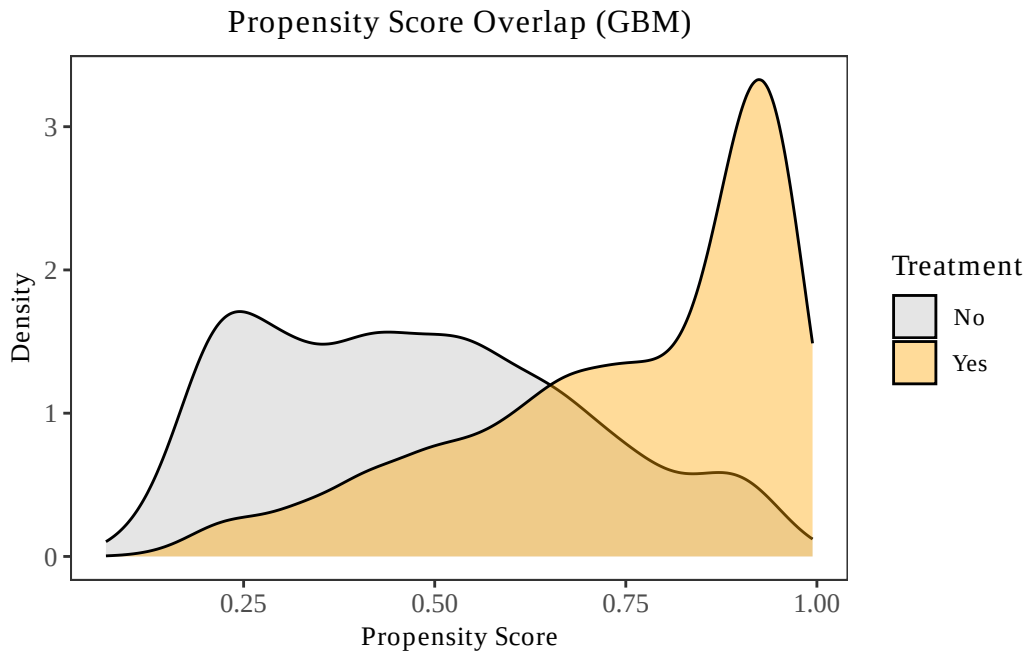


```
## if you want to customize the figure, you may
## manually calculate propensity of being treated/being controlled and plot by ggplot
NCDS$treat_prob <- res_gbm$propensity[,2]
NCDS$control_prob <- res_gbm$propensity[,1]

ggplot(NCDS, aes(x = treat_prob, fill = as.factor(Dany))) +
  geom_density(alpha = 0.4) +
  scale_fill_manual(
    values = c("0" = "grey", "1" = "orange"),      # assign colors to factor levels
    labels = c("0" = "No", "1" = "Yes")           # legend labels
  ) +
  labs(
    fill = "Treatment",
    x = "Propensity Score",
    y = "Density",
    title = "Propensity Score Overlap (GBM)"
  ) +
  theme_bw(base_family = "Palatino") +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_blank(),
    axis.line = element_blank(),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),

```

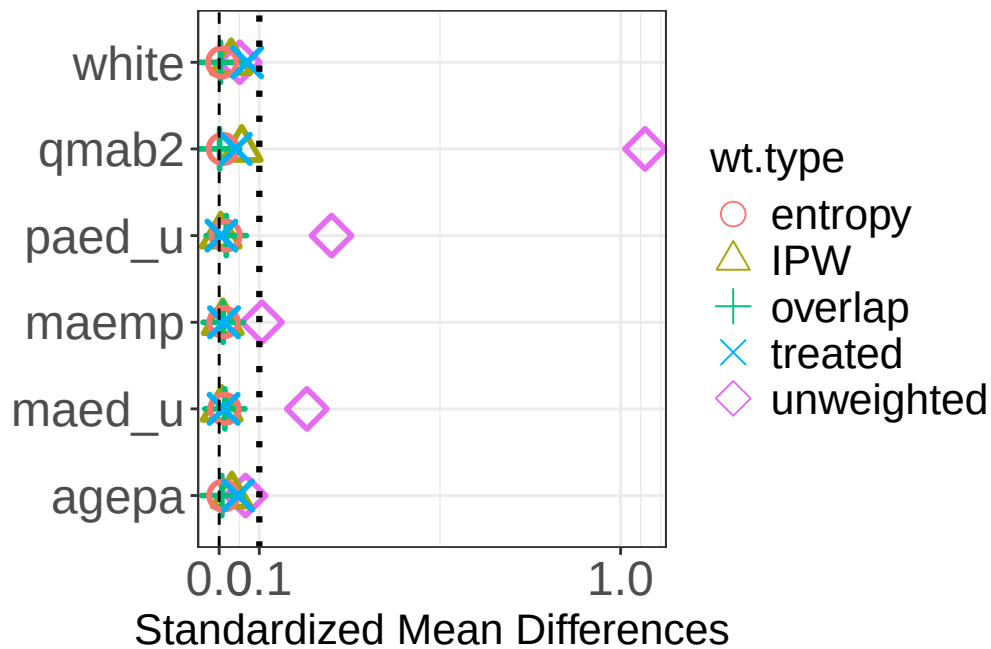
```
axis.title = element_text(size = 10),
plot.title = element_text(size = 12, hjust=0.5)
)
```



Covariate Balance Check

After estimating the propensity scores and constructing weights, it is essential to evaluate covariate balance between treated and control groups. The goal of propensity score weighting is to create a pseudo-population where the distribution of observed covariates is similar across groups—mimicking random assignment. By checking covariate balance, we can verify whether the weighting procedure has successfully reduced systematic differences and minimized confounding bias. Poor balance indicates that the estimated treatment effects may still be biased, while good balance suggests that the weighting has achieved comparability between groups.

```
## PSweight has a built-in function for covariate balance check
plot(bal.mult, metric = "ASD")
```



Estimate the Treatment Effect by Doubly Robust SuperLearner

```
res_dr <- PSweight(
  ps.formula = ps.formula,
  yname = "wage",
  data = NCDS,
  weight = "IPW", ## estimand
  ps.method = "SuperLearner", ## propensity fit by SuperLearner
  augmentation = TRUE,
  out.formula = wage ~ white + maemp + qmab2 + paed_u + maed_u + agepa,
  out.method = "glm", ## this can also be estimated using GLM
)

## summarize results
summary(res_dr)$estimates
```

	Estimate	Std.Error	z value	lwr	upr	Pr(> z)
Contrast 1	0.2399057	0.01605364	14.94401	0.2084412	0.2713703	0

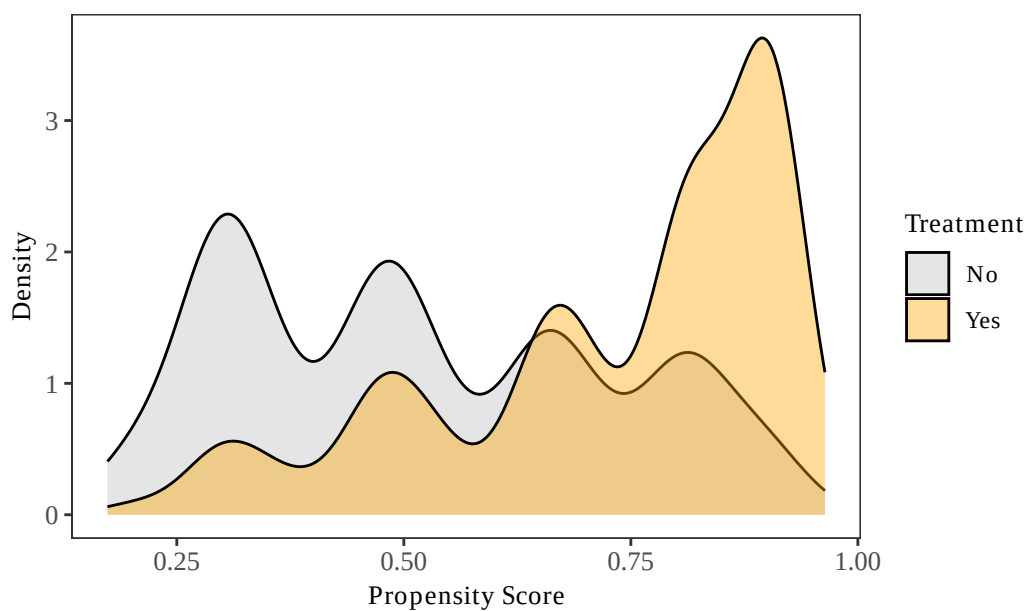
```

## calculate propensity of being treated/being controlled
NCDS$treat_prob <- res_dr$propensity[,2]
NCDS$control_prob <- res_dr$propensity[,1]

ggplot(NCDS, aes(x = treat_prob, fill = as.factor(Dany))) +
  geom_density(alpha = 0.4) +
  scale_fill_manual(
    values = c("0" = "grey", "1" = "orange"),      # assign colors to factor levels
    labels = c("0" = "No", "1" = "Yes")           # legend labels
  ) +
  labs(
    fill = "Treatment",
    x = "Propensity Score",
    y = "Density",
    title = "Propensity Score Overlap (SuperLearner)"
  ) +
  theme_bw(base_family = "Palatino") +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_blank(),
    axis.line = element_blank(),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),
    axis.title = element_text(size = 10),
    title = element_text(size = 10)
  )

```

Propensity Score Overlap (SuperLearner)



```
## check balance
bal.mult <- SumStat(
  ps.formula = ps.formula,
  data = NCDS,
  weight = c("IPW", "overlap", "entropy", "treated"), ## various approaches to weight individuals
  method = "SuperLearner"
)

## plot the propensity score density
plot(bal.mult, metric = "ASD")
```

